Lec-6

08/09/26 2:06 PM

46 — ⓥ

64 — ⑪

2346 → 6432

6 x10+4

N = 2346 → 6432

↓

234

↓

23

↳ 2

↳ 0

Rev = 6 x10+4

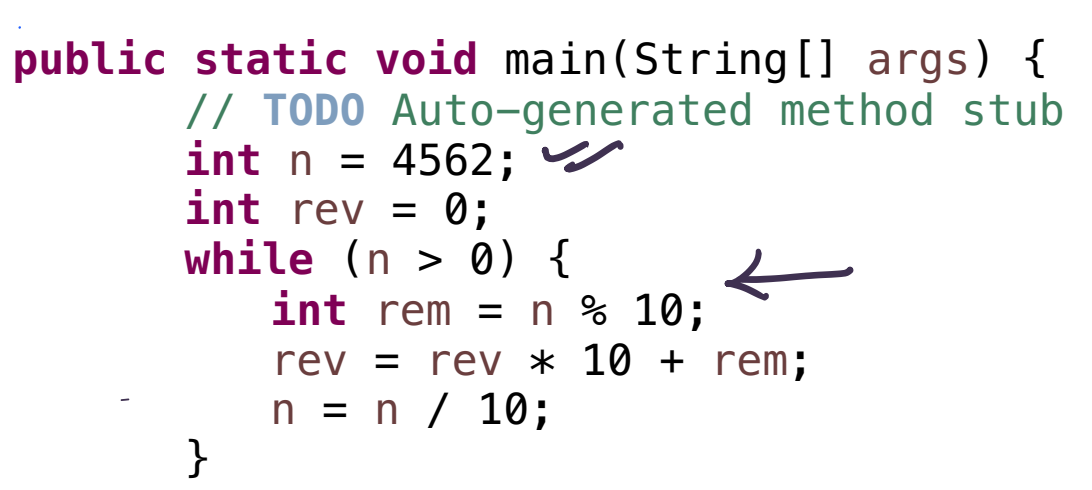
= 64 x10+3

= 643

= 643 x10+2

= 6432

```
while(n>0) {
    rem = n % 10
    Rev = Rev x10 + rem
    n = n/10
}
```



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 4562;
        int rev = 0;
        while (n > 0) {
            int rem = n % 10;
            rev = rev * 10 + rem;
            n = n / 10;
        }
        System.out.println(rev);
    }
```

2654

4562

↓

456

↓

45

↓

4

↓

0

1 1214 121

33

39

rev = 0 x10 + 2

= 2 x10 + 6

= 26 x10 + 5

= 265 x10 + 4

= 2654

4562 →

60, 36

H.c.f →

60 = 1 2 3 4 5 6 10 12 15 20 30 60

36 = 1 2 3 4 6 9 12 18 36

C 1 2 3 4 5 6 (12) → H.c.f

dividend = ~~60~~ ~~36~~ 24

divisor = ~~36~~ ~~24~~ 12

disior ↓ dividend

```
36 ) 60 (
     36
24 ) 36 (
     24
12 ) 24 (
     24
      0
```

12 → H.c.f

```
X while(dividend % divisor != 0) {
      rem = dividend % divisor
      dividend = divisor
      divisor = rem
  }
```

60%36 36%24 24%

```
36 ) 60
     36
24 ) 36 (
     24
12 ) 24
```

L.c.m

a, b

$a \times b = gcd(a,b) \times L.c.m(a,b)$

$L.c.m = \left(\frac{a \times b}{gcd}\right)$

2	378
3	189
3	63
3	21
7	7
	1

2 3 3 3 7

2 x3 x3 x3 x7

2	378
3	189
3	63
3	21
7	7
	1

i = ~~2 3 4 5 6~~ ⑦

```
while(n>1) {
    if(n%i==0) {
        syso(i)
        n=n/i
    }
    else {
        i++
    }
}
```

①

39 ✓ Binary base → 2 ↳ 0,1

✓ octal → base → 8 ↳ 0,7

✓ decimal base → 10 ↳ 0,9

X Hex decimal → base 16 → 0,9 A F

```
  1
  7 8
  6 5
 -----
1 4 3
```

```
 1010
 1101
```

```
  100
  2314 → 5
  4310 → 5
 ------
  12124
```

4+0=4

4%5

4/5=0

2%5=2

2/5=0

6%5=1

6/5=0

9%5=4

9/5 ...

13%10=3

13/10=1

7%10=7

7/10=0

14%10=4

14/10=1

10%10=0

10/10 —

2%2=0

2/1

```
 1 1  1
  3749
  6724
 ------
 10473
```

```
  0111 1
  10110 1
  100111
 --------
 1010100
```

$378 = 3 \times 10^2 + 7 \times 10^1 + 8 \times 10^0$

46

2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
2	1	1
	0	

101110

2	39	1
2	19	1
2	9	1
2	4	0
2	2	0
2	1	1
	0	

100111

$10110 = 1 \times 10^4 + 0 \times 10^3 + 1 \times 10^2 + 1 \times 10^1 + 0 \times 10^0$

```
100000
   100
    10
     1
------
100111
```

sum = 0, mul = ~~1~~ 10 10² 3 4 5

```
while(n>0) {
    rem = n%2
    sum = sum + rem * mul
    n = n/2
    mul = mul x10
}
```

2	39	1×10^0
2	19	1×10
2	9	1×10^2
2	4	0×10^3
2	2	0×10^4
2	1	1×10^5
	0	

$sum = 1 \times 10^0 + 1 \times 10^1 + 1 \times 10^2 + 0 + 0 + 1 \times 10^5$

=